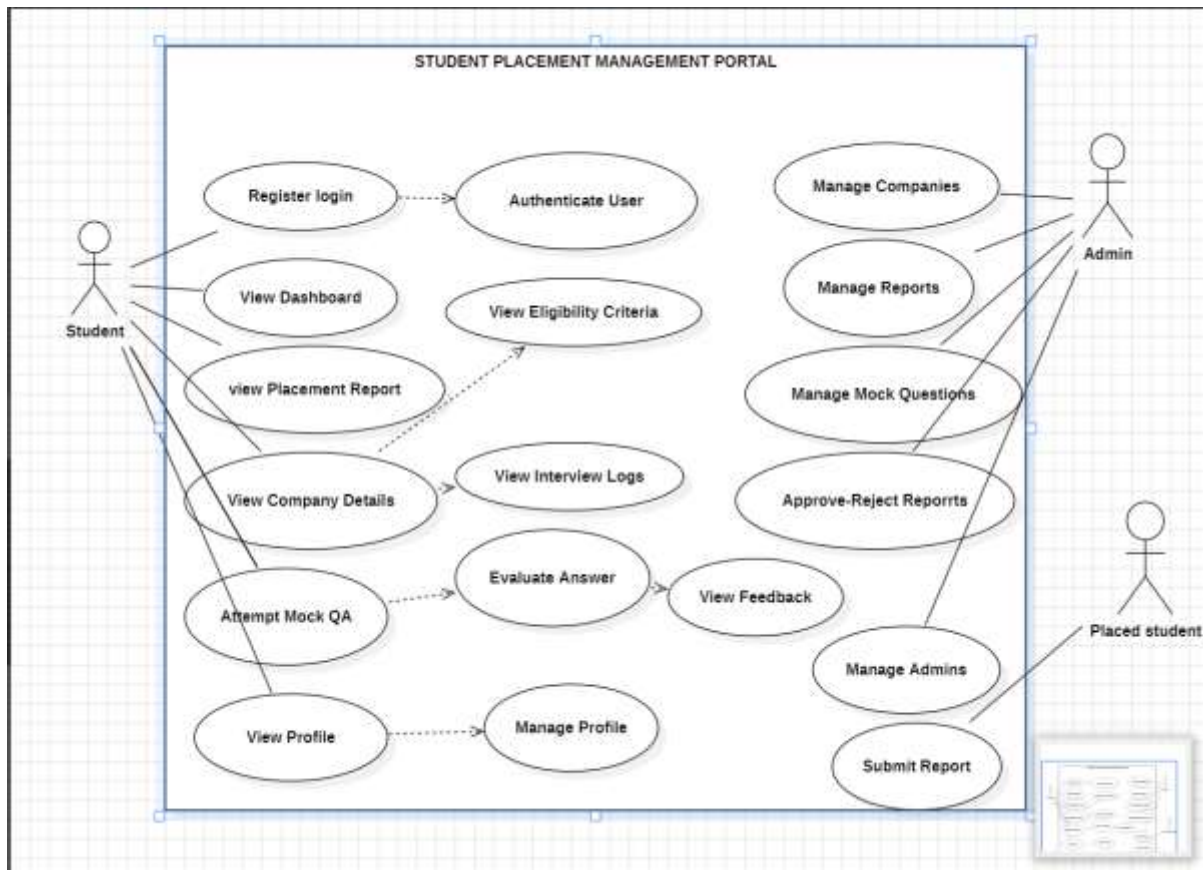




LAB – 5

UML USE CASE MODELING

Project: STUDENT PLACEMENT PORTAL

1. Aim

To identify the actors and use cases of the **Student Placement Portal** from its approved requirements and represent the functional interactions using a UML Use Case Diagram and corresponding use case descriptions.

2. Project Description

The **Student Placement Portal** is a student-centric web platform designed to help students prepare for and participate in campus placement drives. The system provides a centralized platform where students can explore companies, view verified placement reports, check eligibility criteria, study previous interview experiences, and practice Online Assessment (OA) questions.

The portal also maintains detailed placement information such as **company-wise CTC, job roles, eligibility requirements, recruitment rounds, interview experiences and**

placement statistics. This allows students to make better decisions while preparing for different companies.

The system also contains an **Admin Control Center** through which authorized administrators can manage company information, placement reports, mock OA questions and contributed placement experiences. Placed students or seniors can contribute their placement experiences, which can be reviewed and approved by administrators before being displayed to other students.

The application uses a modern single-page interface with dashboard analytics and a multi-layer caching mechanism to improve performance and reduce unnecessary database requests.

3. Actors

1. Student

The Student is the primary actor of the system. Students can:

- Login to the portal
- View placement dashboard
- Browse companies
- View placement reports
- View company details
- Check eligibility criteria
- View interview experiences
- Attempt mock OA questions
- View results and feedback
- Manage their profile

2. Admin

The Admin is responsible for managing the portal and maintaining the quality of placement information. Admin can:

- Manage companies
- Manage placement reports
- Manage mock questions

- Approve or reject submitted reports
- Manage administrator access

3. Placed Student

A placed student/senior can contribute useful placement experiences and reports to the portal. The submitted information is reviewed by the administrator before becoming verified content.

4. Major Use Cases

ID	Use Case	Description
UC-01	Register / Login	Authenticate the user and provide access to the portal.
UC-02	View Dashboard	View placement statistics, companies, CTC and other analytics.
UC-03	View Placement Reports	Browse verified placement experiences and reports.
UC-04	View Company Details	View detailed information about a company and its recruitment process.
UC-05	View Eligibility Criteria	Check the requirements associated with a placement opportunity.
UC-06	View Interview Logs	View round-by-round interview experiences.
UC-07	Attempt Mock OA	Attempt online assessment questions for placement preparation.
UC-08	Evaluate Answers	Evaluate answers submitted during mock practice.
UC-09	View Feedback	View score, correct answers and feedback after an assessment.
UC-10	Manage Profile	View and update relevant student profile information.

ID	Use Case	Description
UC-11	Submit Placement Report	Allow placed students to submit their placement experience.
UC-12	View Submission Status	View the status of a submitted placement report.
UC-13	Manage Companies	Admin can add, update or manage company information.
UC-14	Manage Reports	Admin can maintain placement reports.
UC-15	Manage Mock Questions	Admin can add, update or remove mock OA questions.
UC-16	Approve / Reject Reports	Admin reviews submitted reports and approves or rejects them.

5. Use Case Relationships

The system primarily uses **Association** relationships between actors and their respective use cases.

Some use cases can also be represented using <<include>> relationships where one activity is a required part of another activity.

For example:

Attempt Mock OA

<<include>> → **Evaluate Answers**

and

Evaluate Answers

<<include>> → **View Feedback**

This represents the logical flow of the mock assessment process.

6. Use Case Narratives

UC-01: Register / Login

Actor: Student / Admin / Placed Student

Precondition: User has a valid account or supported authentication method.

Main Flow:

1. User opens the portal.
2. User enters authentication details.
3. System verifies the credentials.
4. System creates an authenticated session.
5. User is redirected to the appropriate portal section.

Alternative Flow:

If the credentials are invalid, the system displays an error message.

Postcondition:

The authenticated user can access the features permitted for their role.

UC-03: View Placement Reports

Actor: Student

Precondition: Verified placement reports are available.

Main Flow:

1. Student opens the Placement Reports section.
2. System retrieves available verified reports.
3. System displays company, year, role, CTC and recruitment information.
4. Student selects a report.
5. System displays detailed placement experience.

Postcondition:

Student can review previous placement experiences and use them for preparation.

UC-05: View Eligibility Criteria

Actor: Student

Precondition: Company eligibility information is available.

Main Flow:

1. Student selects a company.

2. Student opens the eligibility section.
3. System retrieves eligibility requirements.
4. System displays the required academic/profile criteria.

Postcondition:

Student can determine whether the placement opportunity is suitable for them.

UC-07: Attempt Mock OA

Actor: Student

Precondition: Mock OA questions are available.

Main Flow:

1. Student opens Mock OA Practice.
2. Student selects a practice set.
3. System displays the questions.
4. Student submits answers.
5. System records the responses.
6. The answers are sent for evaluation.

Postcondition:

The student's mock assessment attempt is completed and ready for evaluation.

UC-11: Submit Placement Report

Actor: Placed Student

Precondition: The placed student is authenticated.

Main Flow:

1. Student opens the contribution section.
2. Student enters company and placement details.
3. Student enters recruitment-round/interview experience.
4. Student submits the report.
5. System stores the report with a pending review status.
6. Admin reviews the submission.

Postcondition:

The placement report is stored for verification.

UC-16: Approve / Reject Reports

Actor: Admin

Precondition: A placement report is waiting for review.

Main Flow:

1. Admin opens the submitted reports.
2. Admin reviews the placement information.
3. Admin verifies the submitted details.
4. Admin approves or rejects the report.
5. System updates the report status.

Postcondition:

Approved reports become available as verified placement information, while rejected reports remain unavailable to students.

7. Technology Stack

The Student Placement Portal is implemented using the following technologies:

- **Frontend:** React 17
- **Styling:** CSS / SCSS
- **State Management:** Zustand
- **Database:** Supabase PostgreSQL
- **Authentication:** Supabase Authentication / Google OAuth
- **Caching:** Zustand, LocalStorage and Service Worker
- **Application Type:** Single Page Application
- **Serving:** Static Node server

The application follows a **four-layer caching architecture** consisting of in-memory Zustand cache, LocalStorage cache, Service Worker cache and Supabase backend as the source of truth. This improves page loading speed, reduces repeated database requests and provides better offline accessibility.

8. Conclusion

The **Student Placement Portal** provides a centralized platform for students to access placement-related information and prepare for recruitment drives. The UML Use Case Model identifies the **Student, Admin and Placed Student** as the major actors and represents their interactions with the system.

The portal combines placement reports, company information, eligibility criteria, interview experiences, CTC information and mock OA practice into a single system. The use case model provides a clear functional representation of the system and can be used as a foundation for further UML models and system development.
